

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 628 Computational Biology — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Durbin, Eddy, Krogh & Mitchison — Biological Sequence Analysis	The primary text for Modules 02 through 08. Probabilistic models of sequence, done properly, and the HMM chapters are the best treatment anywhere. Library copy.
Jones & Pevzner — An Introduction to Bioinformatics Algorithms	Modules 02, 04, 05, and 08. Algorithm-first, with the biology supplied as needed — which suits this course's audience exactly. Library copy.
Compeau & Pevzner — Bioinformatics Algorithms, with the free Rosalind problems	The exercises for Modules 02 through 08, free. Implementing these is the fastest way to find out whether you understood the recurrence.
Lander & Weinberg and the open genomics literature (free)	Module 01's biology and Modules 10 to 13's practice. PubMed Central is open access and is where the methods sections actually are.
Leek & Storey, and the batch effects literature (free)	Module 10. Why a large fraction of published expression findings were batch structure, and what to do about it.
The Bioconductor and Biopython documentation (free)	Modules 10 and 12. The standard tooling, and Bioconductor's statistical packages encode a great deal of hard-won practice.

Module-by-module map

Module 01 — What Is Actually Measured

Resource	What it covers for this module
The NCBI Science Primer and handbooks (free)	§1 — the biology vocabulary, at the level this course needs and no more.
Scitable and the open genetics primers (free)	§1's exceptions — splicing, overlapping genes, and why 'the gene' is contested.
Illumina and PacBio technology overviews (free)	§2 — what the instruments do, from the manufacturers, read for the mechanism rather than the marketing.
Ioannidis — Why most published research findings are false (free)	§4 and the closing callout — the argument, which is statistical and applies directly to Modules 10 and 11.

Module 02 — Sequence Alignment

Resource	What it covers for this module
Durbin et al., chapter 2 (library copy)	The whole module — alignment developed as probabilistic inference, which makes §1's callout precise.
Needleman & Wunsch (1970), and Smith & Waterman (1981)	§§1 and 2 in the originals — short, and the second is three pages.
Gotoh — An improved algorithm for matching biological sequences	§3's three-state formulation in the original.
Hirschberg — A linear space algorithm for computing maximal common subsequences	§4's divide and conquer — and it predates its biological application entirely.

Module 03 — Scores and Significance

Resource	What it covers for this module
Henikoff & Henikoff — Amino acid substitution matrices from protein blocks (free)	§1's BLOSUM in the original, with the construction explained.
Karlin & Altschul — Methods for assessing the statistical significance of molecular sequence features (free)	§§2 and 3 — where the E-value formula comes from, and the extreme-value argument.
The NCBI BLAST statistics documentation (free)	§3 — how to read a report, by the people who compute the numbers.
Wootton & Federhen — Statistics of local complexity in amino acid sequences	§4's first caveat — low complexity, and why filtering exists.

Module 04 — Search and Indexing

Resource	What it covers for this module
Altschul et al. — Basic local alignment search tool (free)	§1 in the original — and the heuristic's assumption is stated openly in it, which is worth seeing.
Burrows & Wheeler — A block-sorting lossless data compression algorithm	§3 in the original , where it is entirely about compression.
Ferragina & Manzini — Opportunistic data structures with applications	§3's FM-index — the search property, and the space bound that made it matter.
Langmead & Salzberg — Fast gapped-read alignment with Bowtie 2 (free)	§§3 and 4 applied — and the bridge to Module 05.

Module 05 — Mapping and Assembly

Resource	What it covers for this module
Compeau, Pevzner & Tesler — How to apply de Bruijn graphs to genome assembly (free)	§2 — four pages, and the clearest explanation of the reformulation.
Li & Durbin — Fast and accurate short read alignment with Burrows-Wheeler transform (free)	§1 — the mapper most widely used, and its handling of the hard cases.
Nagarajan & Pop — Sequence assembly demystified (free)	§§2 to 4 — including an honest treatment of the metrics.
The Human Pangenome Reference Consortium (free)	§4's closing callout — what replaces a single linear reference, and why.

Module 06 — Multiple Alignment and Phylogeny

Resource	What it covers for this module
Durbin et al., chapters 6 and 7 (library copy)	§§1 and 3 — multiple alignment and phylogeny, probabilistically.
Felsenstein — Inferring Phylogenies	§§3 and 4 — the standard text, and the bootstrap is his. Library copy.
Edgar & Batzoglou — Multiple sequence alignment (free review)	§§1 and 2 — the heuristics compared, with their failure modes.
Felsenstein — Cases in which parsimony or compatibility will be positively misleading	§4's long-branch attraction , in the original — and the title says the finding.

Module 07 — Hidden Markov Models

Resource	What it covers for this module
Durbin et al., chapters 3 to 5 (library copy)	The whole module — and the profile HMM chapters are the definitive treatment.
Rabiner — A tutorial on hidden Markov models (free copies widely available)	§§1 and 2 — the three problems stated clearly, from speech recognition.
Eddy — What is a hidden Markov model? (free)	§1 in two pages — the best short explanation there is.
The HMMER documentation and Pfam (free)	§3 — profile HMMs as working software, and Pfam is the family database built from them.

Module 08 — RNA Structure

Resource	What it covers for this module
Durbin et al., chapter 10 (library copy)	§§2 and 4 — RNA structure, including stochastic context-free grammars, which is the unifying framing.
Zuker — Mfold web server for nucleic acid folding (free)	§2's energy minimisation — and the server is worth running on a sequence you care about.
Lorenz et al. — ViennaRNA Package 2.0 (free)	§§3 and 4 — partition function, base-pair probabilities, and comparative folding, as working software.
Mathews et al. — the nearest-neighbour energy parameters	§2's measured parameters — where the numbers come from, which is laboratory work rather than computation.

Module 09 — Protein Structure

Resource	What it covers for this module
Anfinsen — Principles that govern the folding of protein chains	§1 — the experiment that made the problem well posed.
Marks et al. — Protein structure prediction from sequence variation (free)	§2's coevolution signal , before the deep models — and it is the conceptual core of what followed.
Jumper et al. — Highly accurate protein structure prediction with AlphaFold (free)	§2 — the method paper, and its own limitations section is the model for §3.
The CASP assessments (free)	§§2 and 3 — blind assessment over three decades, which is how the claim was established and is the right evidence to cite.

Module 10 — Expression and Multiple Testing

Resource	What it covers for this module
Love, Huber & Anders — Moderated estimation of fold change and dispersion (DESeq2) (free)	§§1 and 2 — the shrinkage model, explained by its authors, and the software is the standard.
Benjamini & Hochberg — Controlling the false discovery rate	§3 — the original, and the idea is simpler than its reputation.
Leek et al. — Tackling the widespread and critical impact of batch effects (free)	§4 — with worked examples of published findings that were batch structure.
Conesa et al. — A survey of best practices for RNA-seq data analysis (free)	The whole module , as a practical checklist.

Module 11 — Association and Confounding

Resource	What it covers for this module
Visscher et al. — 10 years of GWAS discovery (free)	The whole module — an honest assessment by people in the field, including the overclaiming.
Price et al. — Principal components analysis corrects for stratification (free)	§2's correction , in the original.
Martin et al. — Clinical use of current polygenic risk scores may exacerbate health disparities (free)	§4's transferability problem , measured — and the consequences stated.
Manolio et al. — Finding the missing heritability of complex diseases (free)	§4 — the problem stated, with the candidate explanations laid out.

Module 12 — Pipelines and Reproducibility

Resource	What it covers for this module
Sandve et al. — Ten simple rules for reproducible computational research (free)	§§1 and 3 — two pages, and the rules are the right ones.
The Snakemake and Nextflow documentation (free)	§3's fourth item — and the tutorials are short enough to do in an evening.
The UCSC and Ensembl genome build documentation (free)	§2 — which builds exist, what changed, and how lifting works.
Nature's reporting and code availability policies (free)	§4 — what journals now require, which is a useful minimum checklist even when you are not submitting.

Module 13 — Claiming a Biological Result

Resource	What it covers for this module
Ioannidis — Why most published research findings are false (free)	§1 — the argument, which is about multiple testing and prior probability and applies directly.
Begley & Ioannidis — Reproducibility in science (free)	§2 — what failed to replicate and what the successful analyses had in common.
The ARRIVE and MIAME-style reporting guidelines (free)	§3 — structured reporting checklists, which are a good basis for your own methods section.
Leek & Peng — Statistics: P values are just the tip of the iceberg (free)	§§1 and 3 — the pipeline, not the test, is where the inference happens.